1. **Define the operating system and list the basic services provided by the operating system.**

   Operating System: Software managing computer hardware and facilitating communication between hardware and software.

   <u>Basic Services:</u>

   Process Management

   Memory Management

   File System Management

   Device Management

   Security and Protection

2. **Differentiate among the following types of OS by defining their essential properties.**

a) Distributed system

Software that runs on multiple computers, coordinating their resources and providing a unified interface for users and applications in a networked environment.

b) Real time system

Real-time System: Computer system that processes data and events as they occur, meeting strict timing constraints.

Soft Real-time: Tolerates occasional delays; performance degrades but system remains functional.

Hard Real-time: Requires immediate and guaranteed response; failure to meet deadlines can have severe consequences.

c) Multiprogramming System

Multiple processes would be there in main memory. Manages and executes multiple programs concurrently.

d) Multitasking System

Enables concurrent execution of multiple tasks or processes, allowing users to run and switch between applications seamlessly. E-g Round Robin, time sharing system.

e) Multiprocessing System

Manages and utilizes multiple processors (CPUs) in a computer system, enabling parallel execution of tasks and improving overall system performance.E-g Parallel system.

3. Explain essential features of following structure of O.S

a) Monolithic System

Refers to a software architecture where all components and modules of a system are tightly integrated into a single executable or binary. In a monolithic structure, there is typically little separation between the different functionalities of the system. E-g UNIX

b) Layered Systems

Refers to a software architecture where components or modules are organized into layers, and each layer provides a specific set of services to the layer above it.

c) Micro Kernels

A software architecture where the core functionality of the operating system is minimized, and essential services are implemented as separate, user-space processes. This design aims to keep the kernel small and flexible while moving non-essential functions to user space for better modularity and ease of maintenance.

d) Exokernel

A special way to build an operating system where it lets programs have more direct control over computer hardware, making things faster and flexible. The kernel keeps it simple, and most tasks are done by programs outside of it.

4. **What is the key difference between a trap and an interrupt?**

   Interrupt is a signal emitted by hardware or software when a process event needs attention.

   A mechanism where the hardware or external devices can signal the CPU to temporarily suspend its current execution and handle a specific event.

   Also known as an exception or a software interrupt, it's a mechanism initiated by a program to request a specific service or to handle an error condition.

5. **What are the types of System calls?**

   1. Process Control: The ability to monitor and adjust a process to give a desired output.

   2. File Manipulation: File manipulation functions handle creating, opening, and saving documents.

   3. Device Manipulation: requesting the Devices for acquire the desired output

6. **List any four process management system calls.**

   fork()

   exec()

   wait()

   exit()

7. **Define user mode and kernel mode. Why are two modes required?**

   User mode runs regular applications. It has limited access to system resources.

   Kernel mode is reserved for the OS, it has privileged access to system resources.

   Dual mode provides security and stability. Applications cannot easily enter their critical section. When an application requires special access, it requests the OS to do it in kernel mode.

8. **What are the O.S features required for multiprogramming?**

Process Management

Memory Management

File System Management

Device Management

Security and Protection

9. **What are the advantages and disadvantages of a multiprocessor system?**

   Advantages:

   Increased performance

   Better reliability

   Better response time

   Better user experience

   Disadvantages:

   Increased power consumption

   Complexity

10. **Describe the difference between symmetric and asymmetric multiprocessing?**

    Symmetric:

    All processors in the system are considered equal and can perform identical tasks. There is no designated master processor. Each processor has equal access to the shared main memory.

    Asymmetric:

    There is a designated master processor that controls the system and delegates tasks to subordinate processors. The master processor often has direct control over the system's resources, including memory.

11. **What is the difference between a loosely coupled and tightly coupled system?**

    Multiprocessor system where each processor has its own local memory. The processors communicate via message passing. It’s used in distributed processing systems.

All the processes have a shared memory. They communicate via shared memory. It's used in parallel processing.

12. What are the drawbacks of a monolithic system?

Larger in size.

Difficult debugging and management.

Difficult to add new functionality.

13. Give examples of microkernel.

Mach and L4.

14. What are differences between macro kernel and micro kernel?

<u>Microkernel:</u>

Smaller in size.

Slower.

Easy debugging and management.

Easy to add new functionality.

<u>Macrokernel:</u>

Larger in size.

Faster.

Difficult debugging and management.

Difficult to add new functionality.

15. Explain process control block in details.

process state: the state may be new, ready, and running etc.
program counter: the counter indicates the address of the next instruction to be executed for this process.
CPU register: small, fast storage unit in the CPU used to hold data and instruction being processed and provide quick access to frequently used information, improving the efficiency of CPU operations.
memory management information: memory allocated to the process.

16. What is switching overhead?

Switching overhead is the time it takes for a computer to switch between different processes. Typically during context switching.

17. Explain threads in detail.

Threads are like smaller units within a computer program. Instead of the whole program running as one big piece, it's divided into threads. Each thread can do its own part of the work independently, allowing tasks to be performed simultaneously. Threads within a program share resources like memory but can work on different aspects of the program at the same time, improving overall efficiency.

18. Explain and differentiate between user level and kernel level thread.

User-level threads are managed by user-level libraries, while kernel-level threads are managed by the operating system kernel. User-level threads provide more control to the application, while kernel-level threads offer better integration with the operating system.

19. List the main differences and similarities between threads and process.

Processes are independent programs with their own memory space, while threads are smaller units within a process that share the same memory space.

Both processes and threads can execute concurrently, have their own program counters, and are scheduled by the operating system.

20. What are various criteria for a good process scheduling algorithm?

Maximize CPU utilization.

Maximize throughput.

Minimize turnaround time.

Minimize waiting time.

Minimize response time.

21. Explain any two preemptive scheduling algorithms in brief.

<u>Round-Robin:</u>

Each process is assigned a fixed time slot or quantum. If a process does not complete within its time quantum, it is moved to the back of the queue, and the next process in the queue is given the CPU.

Priority Based:

Each process is assigned a priority value. The process with the highest priority is given the CPU for execution. If a higher-priority process arrives preemption occurs, and the CPU is given to the higher-priority process.

22. Explain the following process scheduling algorithm

a) Multilevel queue:

It organizes the processes in multiple queues with different priority levels. Each queue has its own scheduling algorithm. But it causes starvation.

b) Multilevel feedback queue:

It organizes the processes into multiple queues, allowing processes to dynamically change queues based on their behavior.

c) Shortest job first scheduling:

Is a preemptive scheduling, in SRT, the process with the smallest runtime to complete is scheduled to run next.

23. What is preemptive and nonpreemptive scheduling?

Preemptive:

The OS can interrupt a running process and replace it with another process. Such as SRTF, LRTF, Round-Robin and Priority based.

Non-preemptive:

Once a process starts, it cannot be interrupted until it's done or waiting. This is simpler but can lead to starvation, such as FCFS, SJF, LJF, HRRN, Multilevel queue and priority based.

24. Explain the terms related to IPC –

a) Race condition

When two or more parts of a program try to change shared data at the same time, the final outcome depends on the order of execution, leading to unpredictable results.

b) Critical section

The segment of code where shared data is accessed and possibly manipulated.

c) Semaphores

Semaphores are a synchronization tool with two types −

Binary semaphore can take the value 0 & 1 only.

Counting semaphores can take nonnegative integers values.

Two standard operations, wait and signal are defined on the semaphore. Entry to the critical section is controlled by the wait operation and exit from a critical region is taken care of by signal operation.

```
do {
        Entry section
                Critical section
        Exit section
                Remaining section
 } while (true);
```

25. **What are the 3 requirement solutions to the critical section problem , must satisfy?**

• -Mutual Exclusion:

o Ensures that only one process at a time can execute in its critical section to prevent concurrent access and potential data corruption.

• -Progress:

o Guarantees that if no process is currently in its critical section and some processes are waiting, at least one of them will be granted access, promoting continuous execution.

• -Bounded Waiting:

o Limits the number of times other processes can enter their critical sections before a waiting process is granted access, preventing indefinite postponement of a process.

26. Explain Peterson's solution? Prove that it's correct.

Peterson's solution was proposed to resolve the critical section problem involving only two processes. This solution guarantees that it provides mutual exclusion, bounded waiting, and progress of the processes.

[Pseudo Code]

The correctness is typically established by demonstrating that the solution satisfies three essential conditions,

Mutual exclusion, progress and bounded waiting.

27. Discuss Product-Consumer problem/Bounded Buffer problem.

Producer-Consumer Problem: A classic synchronization problem where two processes, a producer and a consumer, share a common, fixed-size buffer. The producer adds items to the buffer, and the consumer removes items. The challenge is to ensure that the producer does not add items to a full buffer and the consumer does not remove items from an empty buffer, while maintaining proper coordination and avoiding data inconsistency.

28. Dining philosopher problem.

A classic synchronization problem where multiple philosophers sit at a dining table, alternating between thinking and eating. Each philosopher needs both a left and right fork to eat, and the challenge is to design a solution to prevent deadlock and ensure fair access to forks for all philosophers.

29. Readers & writers problem.

A synchronization problem that occurs when multiple processes want to access a shared resource simultaneously. The goal is to allow multiple readers to access the resource at the same time but with only one writer, to avoid conflicting access.

30. Write short on interprocess communication.

Shared Memory: An area of memory shared between processes that wish to communicate.

Message Passing: Processes communicate with each other by establishing a communication link between them and exchanging messages via send/receive function.

31. Define,

a) Shell

A shell is a command-line interface that allows users to interact with the operating system by entering commands.

b) Signal handling

Signal handling is a mechanism in an operating system that allows processes to receive and respond to signals, which are software interrupts or notifications.

d) Starvation

Starvation is a situation in computing where a process is unable to proceed because it is unable to acquire the necessary resources to execute, leading to indefinite delay in its progress.

f) Two level scheduling

Two-level scheduling involves dividing the scheduling process into two levels: a high-level scheduler (long-term scheduler) that selects processes from the job pool to bring into the ready queue, and a low-level scheduler (short-term scheduler) that selects processes from the ready queue to run on the CPU.

32. Explain use of message passing & semaphore for inter process communication?

Message Passing:

Use: Communication between processes.

Mechanism: Explicit message send/receive.

Advantages: Simplicity, supports remote communication.

Semaphore:

Use: Coordinating shared resources.

Mechanism: Shared variable, wait, signal.

Advantages: Mutual exclusion, synchronization.

33. Which are the typical information elements of a file directory?

• Name: only information kept in human-readable form.

• Identifier: unique tag (number) identifies file within file system.

• Type: needed for systems that support different types.

• Location: The location or path where the directory is stored in the file system.

• Size - current file size

34. List 5 file organization.

Heap

Sequential

Hashed

Clustered

Indexed

35. Which are the typical operations performed on a directory?

Create Directory

Delete Directory

Change Directory

Rename Directory

Copy Directory

Move Directory

Set Permissions

36. What are methods of free space management of Disk?

FCFS: first come first serve.

SSTF: shortest seek time first.

Scan: start at one end and move to another, reversing back when it gets to each end.

C-scan (Circular SCAN): instead of reversing it moves from one end to the other, then immediately proceeds back to the end it started scanning from

• Look

37. What is the difference between field and record?

Field: A collection of one or more characters is called a field. It's also called a column.

Record: Collection of fields that represents a complete set of information.

38. Explain multiprogramming with a fixed partition.

Allocates fixed-size partitions in memory to multiple processes, allowing concurrent execution.

39. Explain multiprogramming with dynamic partition.

Allocates variable-sized memory partitions to processes based on their memory requirements, facilitating efficient space utilization.

40. What are the differences between internal and external memory Fragmentation?

<u>Internal Fragmentation:</u> Condition in computer memory management, happens when there is a waste of memory due to the allocator returning a larger block than the memory requested.

<u>External Fragmentation:</u> When the available memory or disk space become scattered or broken into small pieces, making it difficult to find enough space for the new data or program, it happens in the stored system or OS.

41. Explain the following allocation algorithm.

a. First fit b. Best fit c. Worst fit d. Next fit

First fit : allocate the first hole that is big enough.

Best fit : allocate the smallest hole that is big enough.

Worst fit : allocate the largest hole that is big enough.

Next fit : allocate the next hole that is big enough.

42. Explain the difference between logical and physical addresses?

Logical Address: Address generated by the CPU during program execution, also known as a virtual address.

Physical Address: Actual location in the computer's memory hardware where data is stored.

43. Explain paging in detail.

A memory management technique used by OS to efficiently manage the available physical memory by dividing it into small chunks called pages, Only the necessary are loaded into memory when a process needs them , allowing for efficient use of memory.

44. Explain hierarchical page table and inverted page table.

45. What is segmentation?

It is a method in computer systems for dividing a process memory into a smaller part called segment, this helps with better memory utilization, improved protection and clearer structure for the programmers.

46. Use of multiprogramming in memory management.

Efficiency

Balance

Throughput

Responsiveness

47. Write short note on:

a) Not-recently used page replacement algorithm.

(im too tired for this shit man)

b) Optimal page replacement algorithm.

(fr bro)

c) Swapping.

Moving processes temporarily out of memory to a backing store.

d) Relocation and protection.

48. Explain the following page replacement algorithm

i. LRU Least Recently Used

A page replacement algorithm that replaces the page in memory that has not been used for the longest time.

ii. FIFO First in First Out

A page replacement algorithm that replaces the oldest page in memory, based on the order of their arrival.

49. Explain banker's algorithm.

The Banker's Algorithm is a deadlock avoidance algorithm used in resource allocation, ensuring that processes request resources in a way that prevents deadlock by checking if the system remains in a safe state before granting a resource request.

50. Define the following,

a) Arrival Time: The time at which the process enters the ready state.

b) Burst Time: The duration required by the process to get executed by the CPU.

c) Turnaround Time: The total time taken for a process to complete, including both waiting and execution time.

d) Completion Time: The time taken for a process to finish execution.

e) Waiting Time: The total time a process spends waiting in the ready queue before getting CPU time.

f) Response Time: The time elapsed between submitting a request and receiving the first response

“it is what it is”